ML

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→ Machine learning is a subset of AI that helps the computer to learn from data instead of being explicitly programmed. It involves algo that analyses the data, identify patterns and make predictions or decission on new unseen data

→ Hypothesis is a function that the algo uses to make predictions

→ Hypothesis space is a set of all Hypothesis that the current working algo could use based on the training data or algo's structure

→ Inductive bias is a set of all assumptions that the learning algo could used to guide it's ~~learning~~ search within the hypothesis space and make prediction on the unseen data

→ Inductive bias is the prior knowledge or tendency that the algorithms brings to the learning process, influencing which hypothesis should the algo use

→ Training set :- It is the major portion of the data used to train the ML model. It is crucial that the training data set is representative of the overall data distribution to ensure that the model generalize well on unseen data

Test set :- It is minor porsion of the data used to evaluate the model. It is crucial that the test set must be representative of the actual data set. It's purpose is to evaluate the trained model performance on new, unseen data

It is helpfull to identiy :-

To small training dataset
Noisy data
Complex model
Long training time

overfitting : [Performs well on training data and poor on test data]
High variance

underfitting : [performs poor on training data]
High bias

Cross Validation : This is a resampling procedure used to evaluate the performance of ML model and access it's ability to generalize the unseen data. It is a crucial technique for preventing overfitting and obtaing a more relable solution

Bayes Theorem

It is a probablity formula that tells us the probablity for a hypothesis (H) for an evidence (E)

Posterior prob. of event H (hypo)

$$P\left(\frac{H}{E}\right) = \frac{P\left(\frac{E}{H}\right) \cdot P(H)}{P(E)}$$

posterior prob. of event E

Ex $P(\text{Spam}) = 30\% = 0.3$
$P(\text{non-spam}) = 70\% = 0.7$
$P(\text{often}/\text{Spam}) = .8$
$P(\text{often}/\text{not spam}) = .1$

$$P(\text{spam}/\text{often}) = \frac{P(\text{often}/\text{spam}) \times P(\text{spam})}{P(\text{often})}$$

$$= .8 \times .3$$

Note $P(A) = P\left(\frac{A}{B}\right) \cdot P(B) + P\left(\frac{A}{B'}\right) \cdot P(B')$

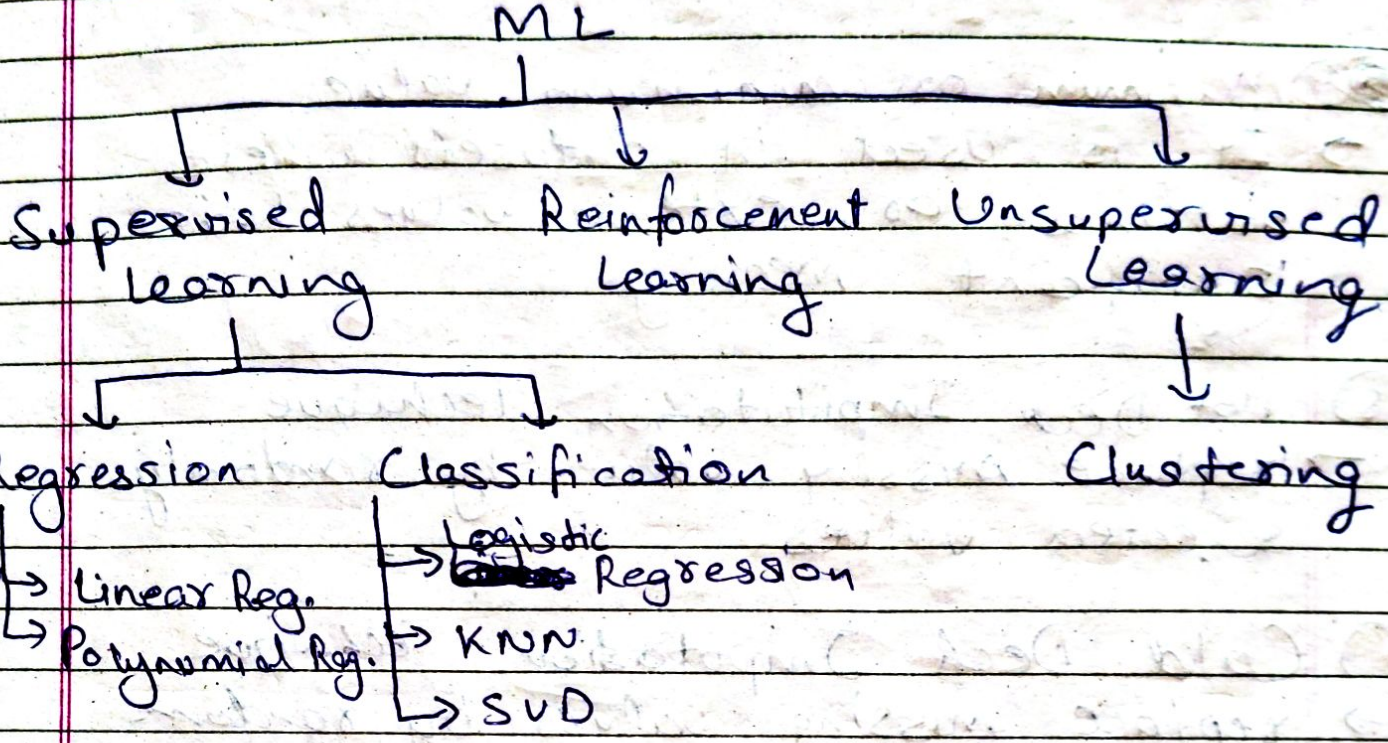
$$P(\text{often}) = P\left(\frac{\text{often}}{\text{spam}}\right) \cdot P(\text{spam}) + P\left(\frac{\text{often}}{\text{non-spam}}\right) \times P(NS)$$

$$P(\text{often}) = .8 \times .3 + .1 \times .7$$
$$= .24 + .07$$
$$= .31$$

$$P\left(\frac{\text{spam}}{\text{often}}\right) = \frac{P\left(\frac{\text{often}}{\text{spam}}\right) \times P(\text{spam})}{P(\text{often})}$$

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$$P\left(\frac{\text{spam}}{\text{offer}}\right) = \frac{.8 \times .3}{.3} = 0.8$$



Data preprocessing

Data imputation: It is the process of fullfilling / replacing the missing values through suitable values so that the data is used by ML algo.

Data imputation techniques :-

① Mean Imputation techniques
→ It can lead to bias, if there is a high percentage of missing values

② Mode Imputation techniques :-
→ This techniques most suitable for catogorical data.

forward fill ← → backward fill

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(3) Next or previous value
→ Use previous or next observed value

(4) Maximum or minimum value
→ It is used in datasets where outlines or extrem values have significant meaning.

(5) Hot Deck Impitutation Technique
→ replace missing value by randomly selected values

(6) Cold Deck Imputation Technique
→ replace missing values by random values from different dataset or historical record.

(7) Regression Imputation Technique
→ It uses regression model to predict missing values
→ The data must follows a linear pattern

(8) K-Nearest Neighbour (KNN) Imputation Techniq
→ finding K-nearest neighbour of each missing value based on the similarity of their non-missing values
→ The missing values are then imputed using the average of K-nearest neighbors.

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→ This technique can handle non-linear relationships b/w features & labels

Outlines: out of the box values

$$z = \frac{x - \mu}{\sigma}$$

$x \rightarrow$ data point
$\mu \rightarrow$ mean
$\sigma \rightarrow$ standard devia

Ex Calculate mean & SD of datapoints

[25, 28, 30, 32, 35, 40, 42, 45, 100, 30, 29]

Mean (μ) = 38.73
SD (σ) = 25.07

$$SD(\sigma) = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \mu)^2}{N}}$$

$$z = \frac{25 - 38.73}{25.07} = -0.55$$

$$z = \frac{125 - 38.73}{25.07} = 3.24$$

The range for outliners is from [-3 to 3]

∴ 25 ≠ outliner
125 ≈ outliner

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Why outliners problematic ?

→ Outliners can skew results, they often misleads models for inaccurate predicts as the models more focuses on outliners in data

① Distort the mean and SD
② Influence regression lines

Identify outliners

→ Use box plot and scatter plot
→ Statistical Methods
    ↳ zscore
    ↳ Interquartile Range (IQR)

→ Handling outliners

① Remove outliners
② Transform data (logarithmic or square root)
③ Use Robust algo
④ Use clipping (bounds)

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Encoding techniques

→ Label encoding
Assiging each catogory a unique numerical representation so that computer can understant and process the data

Cons
Model can set priority based on neumerical da

→ One hot encoding
Assigining each catogory a unique column with 0 (not present) & 1 (present) value

Cons
Create dimensionallity problem (complexity ↑)

Dimensionality Reduction:

Reduction in features without loosing any important information

* Methods

→ Feature selection (select only imp. feature)
  ↳ Correlation analysis
  ↳ Chi-square test
  ↳ Information gain

→ Feature extraction (combining old feature to form new on
  ↳ PCA
  ↳ LDA

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Advantages

- → Reduced Computation
- → Reduced Cost
- → Improved mode performance
- → Better Insights
- → Prevention of overfitting

Application of feature extraction

1 Image processing & computer vision
- → object recognition
- → facial recognition
- → Image classification

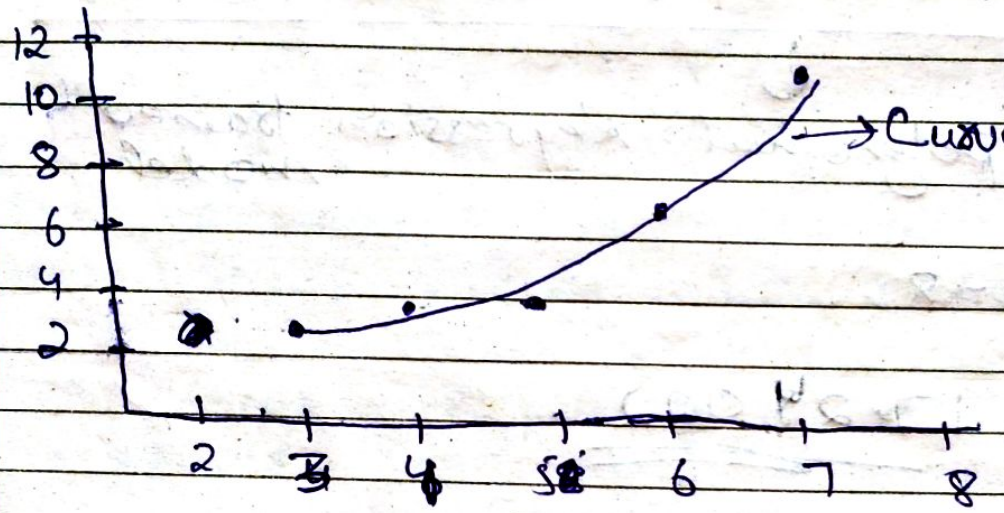
2 Natural Language Processing (NLP)
- → Text classification
- → Sentiment Analysis
- → speech Recognition

3 Biomedical Engineering
- → Medical Image analysis
- → Biological Signal processing
- → Machine condition monitoring

ew one)

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	X	Y	x^2	x^3	x^4	yx	yx^2
	3	2.5	9	27	81	7.5	22.5
	4	3.2	16	64	256	12.8	51.2
	5	3.8	25	125	625	19.0	95
	6	6.5	36	216	1296	39.0	234
	7	11.5	49	343	2401	80.5	563.5
Σ	25	27.5	135	775	1659	158.8	966.2



$$y = a_0 + a_1 x + a_2 x^2$$

$$\sum y_i = n a_0 + a_1 \sum x_i + a_2 \sum x_i^2$$

$$\sum y_i x_i = a_0 \sum x_i + a_1 \sum x_i^2 + a_2 \sum x_i^3$$

$$\sum y_i x_i^2 = a_0 \sum x_i^2 + a_1 \sum x_i^3 + a_2 \sum x_i^4$$

$$27.5 = 5a_0 + 25a_1 + 135a_2 \quad \text{---(I)}$$
$$158.8 = 25a_0 + 135a_1 + 775a_2 \quad \text{---(II)}$$
$$966.2 = 135a_0 + 775a_1 + 1659a_2 \quad \text{---(III)}$$

On solving these eqⁿ

$a_0 = 12.4285$

$a_1 = -5.5128$

$a_2 = 0.76428$

putting these values in eq (11)

$$y = 12.42857 - 5.5128x + 0.76428x^2$$

↓

polynomial Regression trained model.

for $x = 8$

$y = 17.24022$

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Covariance and corelation

covariance quantifies how variables x & Y vary with respect to their means

It is the average product deviated from the mean

$$Cov(x, Y) = \sum (x_i - \bar{x})(y_i - \bar{y})$$

→ +ve covariance indicates as one variable increases, the other also ↑

→ -ve covariance indicates as one variable increases, the other ↓ decreases

⇒ zero covariance indicates that there is no linear relationship b/w two variables

Logistic Regression

It is a supervised ML Algo used to predict binary classification. It predict the probability of an instance belonging to a class using sigmoid function

$$p = \frac{1}{1 + e^{-x}} \rightarrow \text{linear regression}$$

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Correlation

Correlation measures the direction and strength of the linear relationship b/w two variables

$$r = \frac{Cov(x,y)}{\sigma_x \cdot \sigma_y}$$

Q Find corelation & Covariance

$x = 6, 5, 3, 4, 2$

$y = 12, 10, 8, 6, 4$

x	y	$x_i - \bar{x}$	$y_i - \bar{y}$		xy	x^2	y^2
6	12	2	+4	8	72	36	144
5	10	1	2	2	50	25	100
3	8	-1	0	0	24	9	64
4	6	0	-2	0	24	16	36
2	4	-2	-4	8	8	4	16
$\bar{x}=4$							
$\bar{y}=8$	20	40	0	0	18	188	90	360

$$Cov(x,y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n}$$

$$= \frac{18}{5} = 3.6$$

$$r = \frac{Cov(x,y)}{\sigma_x \sigma_y} = \frac{3.6}{4} = 0.9$$

$\sigma \times \sigma$

$$\sigma_x = \sqrt{\frac{\sum (x - \bar{x})^2}{n}} \qquad \sqrt{\frac{10}{5}} \qquad \sqrt{\frac{40}{5}}$$

$\sqrt{2}$ $2\sqrt{2}$

$2 \times 2 = 4$ $\sqrt{8}$

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Confusion Matrix Basis:

	Actual +ve	Actual -ve
pred +ve	TP	FP
pred -ve	FN	TN

~~Accuracy = TP + FP + FN + TN~~

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Precission} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{F1 score} = \frac{2 * (\text{Precision} * \text{Recall})}{\text{Precision} + \text{Recall}}$$

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$$\text{F1-score} = \frac{2 \times P \times R}{P+R} = \frac{2 \times .75 \times .75}{1.5} = .75$$

→ PCA (Principal Component Analysis) is a stastical technique that transforms high-dimensnality data into a smaller set of uncorrelated variables called principal component, while retaining most of the info in data

Q Find eigen values

→ $\bar{x}, \bar{y}$

→ $S = \begin{bmatrix} cov(x,x) & cov(x,y) \\ cov(y,x) & cov(y,y) \end{bmatrix}$

→ $\det |S - \lambda I| = 0$

$a_0 + a_1\lambda + a_2\lambda^2 = C$

$\lambda =$ _____

$\lambda =$ _____ high variance so choose

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K-Nearest Neighbour classifier

KNN is a non parametrized supervised learning classifier which can be used for both predicting regression as well as classification problems

Ecludean distance (d)

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Q $x_1 = 7.5$
$x_2 = 7.5$
$K = 3$

for A $d = \sqrt{(7.5-8)^2 + (7.5-9)^2} = \sqrt{.25 + 2.25} =$

B $d = \sqrt{(7.5-7)^2 + (7.5-8)^2} = \sqrt{.25 + .25} =$

C $d = \sqrt{(7.5-5)^2 + (7.5-7)^2} = \sqrt{6.25 + .25} = 2.$

D $d = \sqrt{(7.5-6)^2 + (7.5-6)^2} = \sqrt{2.25 + 2.25} =$

E $d = \sqrt{(7.5-9)^2 + (7.5-7)^2} = \sqrt{2.25 + .25} =$

B(0.7), A(1.58), E(1.58)

Three closest neighbour

Apple

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→ **Residual**: It is the difference b/w the actual and the predicted value for each datapoint.

→ Represents the error for a specific pred.

$$\text{Residual} = y - \hat{y}$$

actual, pred

→ **MSE (Mean Squared Error)**:

It is the average of squared differences b/w the actual value & predicted value

$$MSE = \frac{\sum (y - \hat{y})^2}{n}$$

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7 Logistic regression is a type of supervised ML. It converts the polynomial regression probability to binary classification with the help of Logistic / Sigmoid function

$$S = \frac{1}{1 + e^{-y}}$$

$\therefore y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_n x_n$ (Linear function)

where, x_n = features like age, BIJ, family history

β_0 are model coefficients

$P \geq 0.5 \longrightarrow 1$

$P < 0.5 \longrightarrow 0$

$$\ln\left(\frac{P}{1-P}\right) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_n x_n$$

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6 ⇒ Linear regression stats a straight line. It is nearly impossible to predict all values through linear reg and it will led to increase in errors

~~Polynomial regression solves this issue by providing bending lines (degree of bending acc to the polynomial deg~~

Polynomial regression provides the linear regression polynomial terms like (x^2, x^3 - -) so it can model non linear relationship b/w feature and labels

$$y = \beta_0 + \beta_1 x^1 + \beta_2 x^2 + \cdots + \beta_n x^n$$

It transforms the input feature x to a polynomial degree like x^2, x^3 - for a better fit of the model but higher degree may led to overfitting (variance) as the model well performs on the training data but purely on test set

→ Use cross-validation to choose polynom
→ Apply regularization
→ prefer low-degree polynomial that balance bias and variance

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outliners $z = \frac{x - \mu}{\sigma}$ (mean, σ → SD)

$\bar{x}, \mu = \frac{\sum x_i}{n}$

SD $(\sigma) = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n}}$

Q_2 (median) $= \left(\frac{n+1}{2}\right)^{th}$ value

$Q_1 = \frac{n+1}{4}$

$Q_3 = \frac{3(n+1)}{4}$

$Q_2 = L + \left(\frac{\frac{N}{2} - cf}{f}\right) \times h$

$COV(X,Y) = \frac{\sum (x - \bar{x})(y - \bar{y})}{N}$

corelation $\leftarrow \gamma = \frac{COV(X,Y)}{\sigma_x \sigma_y}$

$m = \frac{COV(x,y)}{Var(x)}$

$Var(x) = (SD_x)^2$

$P\left(\frac{H}{E}\right) = \frac{P\left(\frac{E}{H}\right) \times P(H)}{P(E)}$

$P(E) = P\left(\frac{E}{H}\right) \cdot P(H^*) + P\left(\frac{E}{H'}\right) \cdot P(H')$